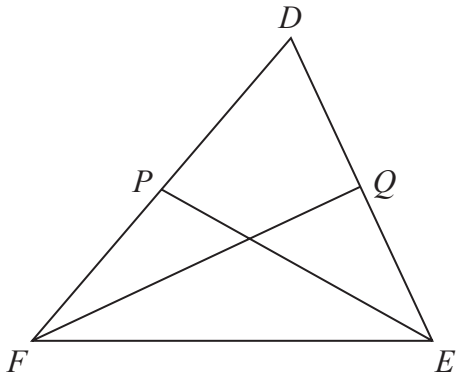
21 *DEF* is a triangle.



P is the midpoint of *FD*.
Q is the midpoint of *DE*.

$\overrightarrow{FD} = \mathbf{a}$ and $\overrightarrow{FE} = \mathbf{b}$

Use a vector method to prove that *PQ* is parallel to *FE*.

DE = FE – FD = b – a; Q is its midpoint.

DQ = 1/2(b – a), and PD = 1/2 a.

PQ = PD + DQ = 1/2 a + 1/2(b – a)

= 1/2 b = 1/2 FE.

Thus PQ is parallel to FE (same vector direction).

4 marks total
DQ = 1/2(b–a)
PQ = PD + DQ
PQ = 1/2 b
Multiple of FE proves parallel

(Total for Question 21 is 4 marks)

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